

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-III

TEST # 05

DATE : 30-09-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 05

Topics

DATE : 30 - 09 - 2018

PRE-MEDICAL NURTURE COURSE : PHASE - III – PATTERN : [NEET]

PHYSICS : Motion in 2D, NLM, Friction

CHEMISTRY : Atomic Structure, Periodic Table, Chemical Bonding, Gaseous State

BIOLOGY : Monera & Protista Kingdom, Animal Kingdom (Ctenophora to Mollusca)

Next Up Coming Test

MINOR TEST # 06

Topics

DATE : 14 - 10 - 2018

PRE-MEDICAL NURTURE COURSE : PHASE - III – PATTERN : [NEET]

PHYSICS : Friction, Circular Motion, Work, Power & Energy

CHEMISTRY : Chemical Bonding, Gaseous State

BIOLOGY : Protista and Fungi Kingdom [Botany], Echinodermata to Agnatha [Zoology]

SPACE FOR ROUGH WORK

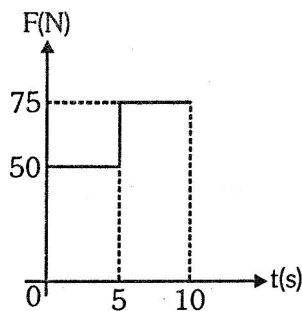
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Three forces act on a particle that moves with unchanging velocity $\vec{v} = (3\hat{i} - 4\hat{j})$ m/s. Two of the forces are $\vec{F}_1 = (3\hat{i} + 2\hat{j} - 4\hat{k})$ N and $\vec{F}_2 = (-5\hat{i} + 8\hat{j} - 3\hat{k})$ N. The third force is :-

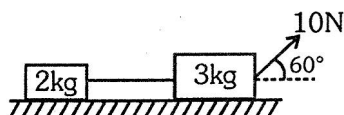
- (1) $(-2\hat{i} + 10\hat{j} - 7\hat{k})$ N
(2) $(2\hat{i} - 10\hat{j} + 7\hat{k})$ N
(3) $(7\hat{i} - 2\hat{k} + 10\hat{j})$ N
(4) none of these

2. A force represented as show in figure acts on a body having a mass of 16 kg. The velocity of the body at $t = 10$ s, if the body starts from rest:-



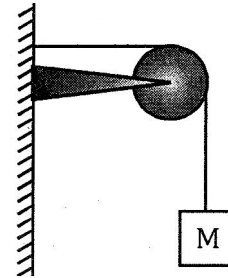
- (1) 100 m/s
(2) 50 m/s
(3) 49 m/s
(4) 39 m/s

3. Find the tension in the string which is connected between the blocks kept on smooth surface as shown in figure :-



- (1) 2 N
(2) 3 N
(3) 5 N
(4) 10 N

4. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :-



- (1) $\sqrt{2} Mg$
(2) $\sqrt{2} mg$
(3) $(\sqrt{(M+m)^2 + m^2})g$
(4) $(\sqrt{(M+m)^2 + M^2})g$

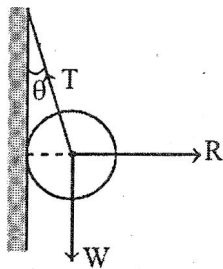
5. A block of mass M is pulled along a horizontal smooth surface by a rope of mass m . Force P is applied at one end of the rope. The force which the rope exert on the block is :-

- (1) $\frac{P}{(M-m)}$
(2) $\frac{PM}{(M+m)}$
(3) $\frac{Pm}{(M+m)}$
(4) $P \frac{m}{M}$

6. A heavy block of mass M is slowly placed on a conveyor belt moving with a speed v . The coefficient of friction between the block and the belt is μ . Through what distance will the block slide on the belt?

- (1) $\frac{v}{\mu g}$ (2) $\frac{v^2}{\mu g}$
(3) $\frac{v}{2\mu g}$ (4) $\frac{v^2}{2\mu g}$

7. When a horse pulls a wagon, the force that causes the horse to move forward is the force
- He exerts on the wagon
 - The wagon exerts on him
 - The ground exerts on him
 - He exerts on the ground
8. A player catches a ball of 200 g moving with a speed of 20 m/s. If the time taken to complete the catch is 0.5 s, the force exerted on the player's hand is :-
- 8 N
 - 4 N
 - 2 N
 - 0
9. If force $F = 500 - 100t$, then impulse as a function of time will be :-
- $500t - 50t^2$
 - $50t - 10$
 - $50 - t^2$
 - $100t^2$
10. A metal sphere of mass 10kg and radius 3m is hung by a string of length 5m fixed to a wall. The forces acting on the sphere are shown in fig. Which of the following represents correct values of forces T and R in Newton.



- 100, 100
 - 125, 75
 - 60, 80
 - 150, 50
11. A block slides with constant velocity on a plane inclined at an angle θ . The same block is pushed up the plane with an initial velocity v_0 . The distance covered by the block before coming to rest is :-

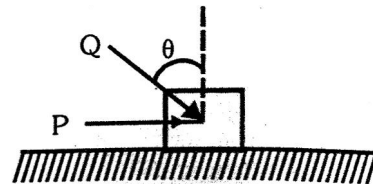
$$(1) \frac{v_0^2}{2g \sin \theta}$$

$$(2) \frac{v_0^2}{4g \sin \theta}$$

$$(3) \frac{v_0^2 \sin^2 \theta}{2g}$$

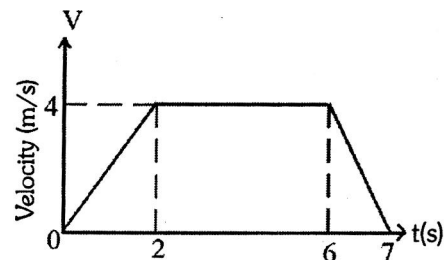
$$(4) \frac{v_0^2 \sin^2 \theta}{4g}$$

12. A block of mass m lying on a rough horizontal plane is acted upon by a horizontal force P and another force Q inclined at an angle θ to the vertical. The block will remain in equilibrium if the coefficient of friction between it and the surface is :-



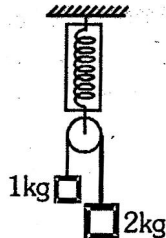
- $\frac{P + Q \sin \theta}{mg + Q \cos \theta}$
- $\frac{P \cos \theta + Q}{mg - Q \sin \theta}$
- $\frac{P + Q \cos \theta}{mg + Q \sin \theta}$
- $\frac{P \sin \theta + Q}{mg - Q \cos \theta}$

13. A lift of mass 100 kg starts moving from rests in the upward direction. Fig shows the variation if speed of the lift, T_1 , T_2 and T_3 stand for tension in the rope from zero to two seconds, two to six seconds, six to seven seconds respectively then Which of the following holds true:-

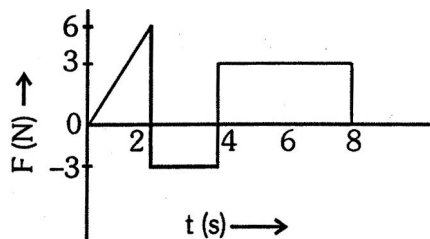


- $T_1 = T_2 = T_3$
- $T_1 > T_2 > T_3$
- $T_1 < T_2 < T_3$
- $T_1 > T_2 < T_3$

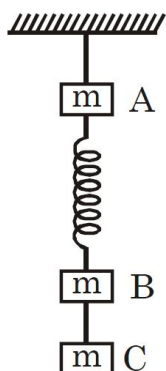
14. Two masses of 1 kg and 2 kg are attached to the ends of a massless string passing over a pulley of negligible weight. The pulley itself is attached to a light spring balance as shown in figure. The masses start moving; during this interval the reading of spring balance will be:-



- (1) More than 3 kg.
(2) Less than 3 kg.
(3) Equal to 3 kg.
(4) None of the above
15. The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-

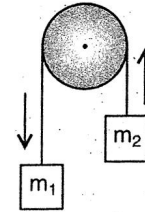


- (1) 24 Ns
(2) 20 Ns
(3) 12 Ns
(4) 6 Ns
16. In the figure shown, find the accelerations of blocks A, B and C immediately after string connecting block A with ceiling is cut :-



- (1) $a_A = 3g\downarrow$, $a_B = g\downarrow$, $a_C = g\downarrow$
(2) $a_A = 2g\downarrow$, $a_B = a_C = 0$
(3) $a_A = 3g\downarrow$, $a_B = a_C = 0$
(4) $a_A = 3g\downarrow$, $a_B = 0$, $a_C = g\downarrow$

17. A massless string passes over a frictionless pulley and carries two masses m_1 and m_2 ($m_1 > m_2$) at its ends. If g is the acceleration due to gravity, then the thrust on the pulley is: ($m_1 > m_2$)



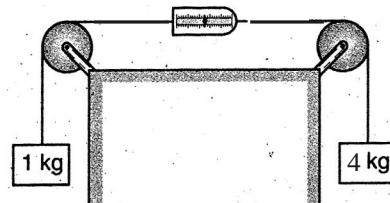
(1) $\left(\frac{2m_1 m_2}{m_1 - m_2} \right) g$

(2) $\left(\frac{2m_1 m_2}{m_1 + m_2} \right) g$

(3) $\left(\frac{4m_1 m_2}{m_1 + m_2} \right) g$

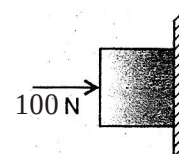
(4) zero

18. In the figure given below, what is the reading of the balance? (Take $g = 10 \text{ N kg}^{-1}$)



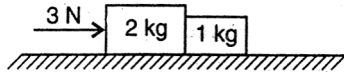
- (1) 24 N
(2) 40 N
(3) 16 N
(4) zero

19. A horizontal force of 100 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is:

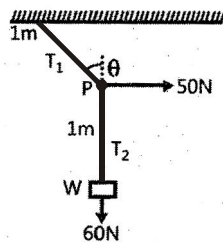


- (1) 2 N
(2) 20 N
(3) 50 N
(4) 100 N

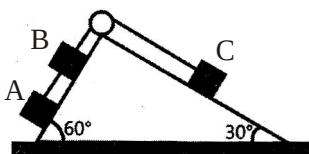
20. Two blocks of 2 kg and 1 kg are in contact on a frictionless table. If a force of 3 N is applied on 2 kg block, then the force of contact between the two blocks will be:



- (1) 0 N (2) 1 N
(3) 2N (4) 3 N
21. In fig. a mass of 6 kg is suspended by a rope of length 2 m from the ceiling. A force of 50N in the horizontal direction is applied at midpoint P of the rope, as shown. What is the angle the rope makes with the vertical in equilibrium? (take g Neglect mass of the rope.)

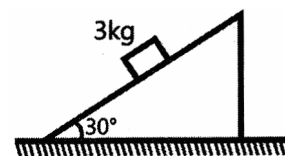


- (1) $\theta = \tan^{-1}\left(\frac{5}{6}\right)$
(2) $\theta = \tan^{-1}\left(\frac{4}{5}\right)$
(3) $\theta = \tan^{-1}\left(\frac{3}{4}\right)$
(4) $\theta = \tan^{-1}\left(\frac{6}{5}\right)$
22. In Fig. masses A, B and C are 1 kg, 2 kg and 3 kg, respectively. Find the acceleration of the system :-



- (1) zero (2) $\frac{g}{6}$
(3) 3g (4) $\frac{g}{4}(\sqrt{3}-1)$

23. A projectile is projected with an initial velocity of $(6\hat{i} + 8\hat{j})\text{ms}^{-1}$ then calculate its horizontal range, maximum height.
(1) 8.6 m, 4.2 m
(2) 9.6 m, 3.2 m
(3) 7.6 m, 6.2 m
(4) 4.6 m, 5.2 m
24. Particle is dropped from the height of 20 m from horizontal ground. A constant force acts on the particle in horizontal direction due to which horizontal acceleration of the particle becomes 6 ms^{-2} . Find the horizontal displacement of the particle till it reaches ground.
(1) 6m (2) 10m
(3) 12m (4) 24m
25. A stone is thrown horizontally from a tower. In 0.5 second after the stone began to move, the numerical value of its velocity was 1.5 times its initial velocity. Find the initial velocity of stone.
(1) 4.4 m/s (2) 4 m/s
(3) 5 m/s (4) 5.4 m/s
26. A boat having a speed of 5 km/hr. in still water, crossed a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in Km/hr.
(1) 1 (2) 3
(3) 4 (4) $\sqrt{41}$
27. A block of mass 3kg is at rest on a rough inclined plane as shown in the Fig.. The magnitude of net force exerted by the surface on the block will be

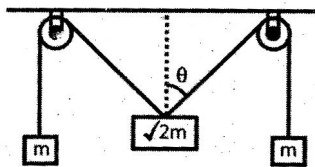


- (1) 26N (2) 19.5N
(3) 10N (4) 30N

28. A block is kept on the floor of an elevator at rest. The elevator starts descending with an acceleration of 12m/s^2 . Find the displacement (in meter) of the block during the first 2 s after the start. Take $g=10\text{m/s}^2$.

(1) 1 (2) 2
(3) 3 (4) 4

29. The pulleys and strings shown in the Fig. are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be

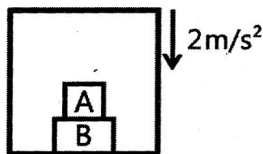


(1) 0° (2) 30° (3) 45° (4) 60°

30. Which of the following represents 2nd law of motion most correctly.

(1) $\vec{F} = m\vec{a}$ (2) $\vec{F} = m \frac{d\vec{v}}{dt}$
(3) $\vec{F} = \frac{d\vec{p}}{dt}$ (4) $\vec{F} = m\vec{v}$

31. The elevator shown in the Fig. is descending with an acceleration of 2m/s^2 . The mass of block A is 0.5 kg. What force exerted by the block A on the block B?



(1) $N = 4\text{ N}$ (2) $N = 5\text{ N}$
(3) $N = 5.5\text{ N}$ (4) $N = 4.5\text{ N}$

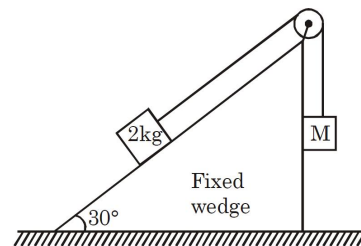
32. A rope of length L is pulled by a constant force F . What is the tension in the rope at a distance x from one end where the force is applied?

(1) $\frac{FL}{L-x}$ (2) $T = \frac{F(L-x)}{L}$
(3) Fx (4) $\frac{FL}{x}$

33. A lift is going upward when suddenly cable supporting the lift breaks. From a reference frame attached to the lift, pseudo force acting on an object will have direction (After the cable broke) :-

(1) Vertically downward when lift goes upward and vertically upward when lift goes downward.
(2) Always vertically upward
(3) Always vertically downward
(4) Vertically upward when lift goes upward and vertically downward when lift goes downward.

34. In the figure shown, find the value of M , for which system is in equilibrium (Neglect mass of string and friction every where).



(1) 10 kg (2) 10.5 kg
(3) 2 kg (4) 1 kg

35. The equation of a projectile is $y = ax - bx^2$. Its horizontal range is :-

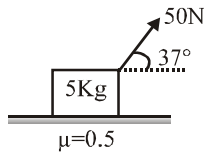
(1) $\frac{a^2}{2b}$ (2) $\frac{a^2}{4b}$
(3) $\frac{a}{b}$ (4) $\frac{b}{a}$

36. The friction force acting on the block at time $t = 4\text{ sec}$ will be :-



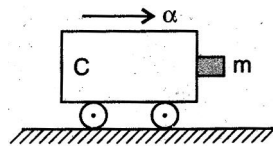
(1) 50 N (2) 30 N
(3) 25 N (4) None

37. The acceleration of the block is :-



- (1) 6 m/s^2 (2) 8 m/s^2
(3) 10 m/s^2 (4) 12 m/s^2

38. A block of mass m is in contact with the cart C as shown in the figure. The coefficient of static friction between the block and the cart is μ . The acceleration α of the cart that will prevent the blocks from falling satisfies:



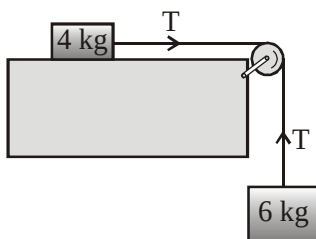
- (1) $\alpha > \frac{mg}{\mu}$ (2) $\alpha > \frac{g}{\mu m}$
(3) $\alpha \geq \frac{g}{\mu}$ (4) $\alpha < \frac{g}{\mu}$

39. A block of mass 2 kg is placed on the floor. The coefficient of static friction is 0.4. A force of 2.8 N is applied on the block as shown in figure. The force of friction between the block and the floor is : (Take $g = 10 \text{ m/s}^2$)



- (1) 2.8 N (2) 8 N
(3) 20 N (4) zero

40. Two bodies of mass 6 kg and 4 kg are tied to a string as shown in the adjoining figure. If the table is smooth and pulley frictionless, then acceleration of mass 6 kg will be : ($g = 10 \text{ m/s}^2$)



- (1) 60 m/s^2 (2) 40 m/s^2
(3) 6 m/s^2 (4) 4 m/s^2

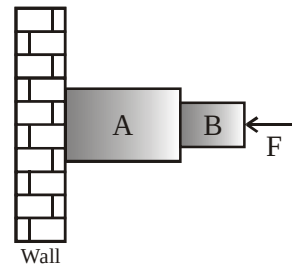
41. With what minimum acceleration can a fireman slide down a rope whose breaking strength is $\frac{2}{5}$ th of his weight?

- (1) 1 g (2) 0.4 g
(3) 0.6 g (4) 0.8 g

42. A diwali rocket is ejecting 0.05 kg/sec of gases at a velocity of 400 m/s. The force on the rocket will be :-

- (1) 22 dyne (2) 20 N
(3) 20 dyne (4) 100 N

43. Adjoining figure shows two blocks A and B pushed against the wall with the force F. The wall is smooth but the surfaces in contact of A and B are rough. Which of the following is true for the system of blocks to be at rest against the wall?



- (1) F should be more than the weight of A and B
(2) F should be equal to the weight of A and B
(3) F should be less than the weight of A and B
(4) System cannot be in equilibrium

44. A block B is pushed momentarily along a horizontal surface with an initial velocity V. If μ is the coefficient of sliding friction between B and the surface, block B will come to rest after a time :-



- (1) $g\mu/V$ (2) g/V
(3) V/g (4) $V/(g\mu)$

45. In case of projectile motion if two projectiles A and B are projected with same speed at angle 15° and 75° respectively to the horizontal then :-

- (1) $R_A > R_B$ (2) $H_A < H_B$
(3) $T_A > T_B$ (4) $T_A = T_B$

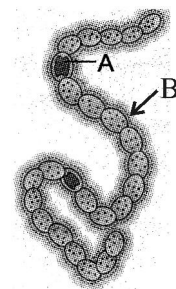
46. Hydrogen by donating one electron forms H^+ . In this property, it resembles with
 (1) Transitional metals
 (2) Alkaline earth metals
 (3) Alkali metals
 (4) Halogens
47. On moving from left to right across a period in the table the metallic character
 (1) Increases
 (2) Decreases
 (3) Remains constant
 (4) First increases and then decreases
48. The electronic structure $(n-1)d^{1-10}ns^{0-2}$ is characteristic of
 (1) Transition elements
 (2) Lanthanides
 (3) Actinides
 (4) Rare gases
49. If three unreactive gases having partial pressures P_A , P_B and P_C and their moles are 1, 2 and 3 respectively then their total pressure will be
 (1) $P = P_A + P_B + P_C$ (2) $P = \frac{P_A + P_B + P_C}{6}$
 (3) $P = \frac{\sqrt{P_A + P_B + P_C}}{3}$ (4) None
50. The correct order of radii is
 (1) $N < Be < B$
 (2) $F^- < O^{2-} < N^{3-}$
 (3) $Na < Li < K$
 (4) $Fe^{3+} < Fe^{2+} < Fe^{4+}$
51. A gas diffuse 1/5 times as fast as hydrogen. Its molecular weight is
 (1) 50 (2) 25
 (3) $25\sqrt{2}$ (4) $50\sqrt{2}$
52. The first ionization potentials (eV) of Be and B respectively are
 (1) 8.29 eV, 9.32 eV
 (2) 9.32 eV, 9.32 eV
 (3) 8.29 eV, 8.29 eV
 (4) 9.32 eV, 8.29 eV
53. $A \rightarrow A^+ + e$, E_1 and $A^+ \rightarrow A^{2+} + e$, E_2 . The energy required to pull out the two electrons are E_1 and E_2 respectively. The correct relationship between two energy would be
 (1) $E_1 < E_2$ (2) $E_1 = E_2$
 (3) $E_1 > E_2$ (4) $E_1 \neq E_2$
54. The electron affinity values for the halogens show the following trend
 (1) $F < Cl < Br < I$
 (2) $F < Cl < Br < I$
 (3) $F > Cl > Br > I$
 (4) $F < Cl > Br > I$
55. Nitrogen has lower electron affinity than its preceding element carbon because
 (1) Electron affinity decreases along a period
 (2) Electron affinity generally increases along a period
 (3) Nitrogen atom has half filled p -orbital
 (4) Nitrogen is a p -block element
56. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 the correct order of acid strength is
 (1) $Al_2O_3 < SiO_2 < SO_2 < P_2O_3$
 (2) $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$
 (3) $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$
 (4) $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$
57. The correct order of the increasing ionic character is
 (1) $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$
 (2) $BeCl_2 < MgCl_2 < BaCl_2 > CaCl_2$
 (3) $BeCl_2 < BaCl_2 < MgCl_2 < CaCl_2$
 (4) $BaCl_2 < CaCl_2 < MgCl_2 < BeCl_2$
58. Increasing order of acid strength of halogen acid is
 (1) $HF < HCl < HBr < HI$
 (2) $HCl < HBr < HI < HF$
 (3) $HF < HI < HBr < HCl$
 (4) None of these
59. The pair of amphoteric hydroxides is
 (1) $Al(OH)_3$, $LiOH$
 (2) $Be(OH)_2$, $Mg(OH)_2$
 (3) $B(OH)_3$, $Be(OH)_2$
 (4) $Be(OH)_2$, $Zn(OH)_2$

60. Rutherford's alpha particle scattering experiment eventually led to the conclusion that
- Mass and energy are related
 - Electrons occupy space around the nucleus
 - Neutrons are buried deep in the nucleus
 - The point of impact with matter can be precisely determined
61. If rate of diffusion of A is 5 times that of B, what will be the density ratio of A and B
- 1/25
 - 1/5
 - 25
 - 4
62. When an electron jumps from L to K shell
- Energy is absorbed
 - Energy is released
 - Energy is sometimes absorbed and sometimes released
 - Energy is neither absorbed nor released
63. Electron occupies the available orbital singly before pairing in any one orbital occurs, it is
- Pauli's exclusion principle
 - Hund's Rule
 - Heisenberg's principle
 - Prout's hypothesis
64. The energy of a radiation of wavelength 8000 Å is E_1 and energy of a radiation of wavelength 16000 Å is E_2 . What is the relation between these two
- $E_1 = 6E_2$
 - $E_1 = 2E_2$
 - $E_1 = 4E_2$
 - $E_1 = 1/2E_2$
65. The energy of electron in hydrogen atom in its ground state is -13.6 eV. The energy of the level corresponding to the quantum number equal to 5 is
- 0.54 eV
 - 0.85 eV
 - 0.64 eV
 - 0.40 eV
66. The uncertainty in momentum of an electron is 1×10^{-5} kg m/s. The uncertainty in its position will be ($h = 6.63 \times 10^{-34}$ Js)
- 5.28×10^{-30} m
 - 5.25×10^{-28} m
 - 1.05×10^{-26} m
 - 2.715×10^{-30} m
67. Which of the following is **INCORRECT** order :-
- LiF > NaF (Lattice energy)
 - LiF > NaF (Inter nuclear distance)
 - $\text{Li}^+ > \text{Na}^+$ (Hydration energy)
 - $\text{Li}^+ < \text{Na}^+$ (Ionic mobility in water)
68. According to Fajan's rule, covalent bond is favoured by
- Large cation and small anion
 - Large cation and large anion
 - Small cation and large anion
 - Small cation and small anion
69. Amongst LiCl, RbCl, BeCl_2 and MgCl_2 the compounds with the greatest and the least ionic character, respectively, are
- LiCl and RbCl
 - RbCl and BeCl_2
 - RbCl and MgCl_2
 - MgCl_2 and BeCl_2
70. The hybridization of IF_7 is
- sp^3d^3
 - sp^2d
 - d^2sp^3
 - sp^3
71. As the s-character of hybridisation orbital increases, the bond angle
- Increases
 - Decreases
 - Becomes zero
 - Does not change
72. Which has sp^3 hybridization of central atom
- PCl_3
 - SO_3
 - BF_3
 - NO_3^-
73. The pair having similar geometry is
- PCl_5 , NH_3
 - BeCl_2 , H_2O
 - CH_4 , CCl_4
 - IF_5 , PF_5
74. The d-orbital involved in sp^3d hybridisation is
- $\text{d}_{x^2-y^2}$
 - d_{xy}
 - d_{z^2}
 - d_{zx}

75. Resonance is due to
 (1) Delocalization of σ electrons
 (2) Delocalization of π electrons
 (3) Migration of H atoms
 (4) Migration of protons
76. The molecular shapes of SF_4 , CF_4 and XeF_4 are
 (1) The same with 2, 0 and 1 lone pairs of electrons on the central atom, respectively
 (2) The same with 1, 1 and 1 lone pair of electrons on the central atoms, respectively
 (3) Different with 0, 1 and 2 lone pairs of electrons on the central atom, respectively
 (4) Different with 1, 0 and 2 lone pairs of electrons on the central atom, respectively
77. Containers A and B have same gases. Pressure, volume and temperature of A are all twice that of B, then the ratio of number of molecules of A and B are
 (1) 1 : 2 (2) 2
 (3) 1 : 4 (4) 4
78. 4 gm of sulphur dioxide gas diffuses from a container in 8 min. Mass of helium gas diffusing from the same container over the same time interval is :-
 (1) 1 g (2) 2 g (3) 3 g (4) 4 g
79. Which one of the following species is diamagnetic in nature
 (1) He_2^+ (2) H_2
 (3) H_2^+ (4) H_2^-
80. Which one is paramagnetic and has the bond order $1/2$
 (1) O_2 (2) N_2 (3) F_2 (4) H_2^+
81. According to kinetic theory of gases,
 (1) There are intermolecular attractions
 (2) Molecules have considerable volume
 (3) No intermolecular attractions
 (4) The velocity of molecules decreases after each collision
82. In which of the following pairs the two molecules have identical bond order
 (1) $\text{N}_2, \text{O}_2^{2+}$ (2) N_2, O_2^-
 (3) N_2^-, O_2 (4) O_2^+, N_2
83. In O_2^- , O_2 and O_2^{2-} molecular species, the total number of antibonding electrons respectively are
 (1) 7, 6, 8 (2) 1, 0, 2
 (3) 6, 6, 6 (4) 8, 6, 8
84. The correct order in which the O – O bond length increases in the following is
 (1) $\text{H}_2\text{O}_2 < \text{O}_2 < \text{O}_3$ (2) $\text{O}_2 < \text{H}_2\text{O}_2 < \text{O}_3$
 (3) $\text{O}_2 < \text{O}_3 < \text{H}_2\text{O}_2$ (4) $\text{O}_3 < \text{H}_2\text{O}_2 < \text{O}_2$
85. Which contains strongest H-bond
 (1) $\text{O}-\text{H} \cdots \text{S}$ (2) $\text{S}-\text{H} \cdots \text{O}$
 (3) $\text{F}-\text{H} \cdots \text{F}$ (4) $\text{F}-\text{H} \cdots \text{O}$
86. The high boiling point of ethanol (78.2°C) compared to dimethyl ether (-23.6°C), though both having the same molecular formulae $\text{C}_2\text{H}_6\text{O}$, is due to
 (1) Hydrogen bonding
 (2) Ionic bonding
 (3) Coordinate covalent bonding
 (4) Resonance
87. Lattice energy of alkali metal chlorides follows the order
 (1) $\text{LiCl} > \text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl}$
 (2) $\text{CsCl} > \text{NaCl} > \text{KCl} > \text{RbCl} > \text{LiCl}$
 (3) $\text{LiCl} > \text{CsCl} > \text{NaCl} > \text{KCl} > \text{RbCl}$
 (4) $\text{NaCl} > \text{LiCl} > \text{KCl} > \text{RbCl} > \text{CsCl}$
88. The correct order of increasing C–O bond length of CO , CO_3^{2-} , CO_2 is
 (1) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
 (2) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
 (3) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$
 (4) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
89. The hybridization of Xe in XeO_2F_2 and its shape is:
 (1) sp^3 , tetrahedral
 (2) sp^3d , TBP
 (3) sp^3d , see – saw
 (4) sp^3d^2 , square Pyramidal
90. The rms velocity at NTP of the species can be calculated from the expression
 (1) $\sqrt{\frac{3P}{d}}$ (2) $\sqrt{\frac{3PV}{M}}$
 (3) $\sqrt{\frac{3RT}{M}}$ (4) All the above

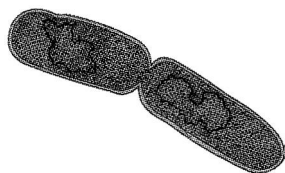
91. The main difference between gram positive and gram negative bacteria in the composition of
- (1) Cilia
 - (2) Cell wall
 - (3) Cell membrane
 - (4) Cytoplasm
92. Moneran lack of cell wall are
- (1) Eubacteria
 - (2) Chlamydia
 - (3) Mycoplasma
 - (4) Cyanobacteria
93. Mycoplasma are smallest living cell known and can survive without
- (1) Cell membrane
 - (2) Oxygen
 - (3) Heterotrophic nutrition
 - (4) Genetic material
94. Which type of genetic material found in bacteria?
- (1) Double stranded circular DNA without histone protein
 - (2) Double stranded linear DNA without histone protein
 - (3) Single stranded DNA with high GC content
 - (4) Low molecular weight single stranded RNA
95. *Rhodospirillum* is a
- (1) Purple sulphur bacteria
 - (2) Purple nonsulphur bacteria
 - (3) Green sulphur bacteria
 - (4) Green nonsulphur bacteria
96. Purple sulphur bacteria
- (1) Are photo organotroph
 - (2) Possess chlorophyll a and phycobillins
 - (3) Use H_2S as H^+ donor
 - (4) Perform oxygenic photosynthesis
97. *E. Coli* and *Anabaena* resemble in the presence of
- (1) Heterocyst
 - (2) Oxygenic photosynthesis
 - (3) Peptidoglycan cell wall
 - (4) Chromatophore

98. The process of genetic recombination in which DNA is transferred from a bacterium to another bacterium by physical contact is known as
- (1) Transformation
 - (2) Conjugation
 - (3) Specialised transduction
 - (4) Generalised transduction
99. Identify the given diagram with correct label.



- (1) Nostoc A-Heterocyst B-Cell wall
 - (2) Nostoc A-Heterocyst B-Mucilaginous sheath
 - (3) Nostoc A-Vegetative cell B-Cell wall
 - (4) Actinomycetes A-Endospore B-Glycocalyx
100. If a bacterium grow only in anaerobic environment and die in the presence of oxygen then it is called as
- (1) Obligate aerobe
 - (2) Facultative aerobe
 - (3) Obligate anaerobe
 - (4) Facultative anaerobe
101. Bacteria reproduce mainly by
- (1) Binary fission
 - (2) Multiple fission
 - (3) Budding
 - (4) Parasexual reproduction
102. During transduction in bacteria, a small segment of double stranded DNA is transferred from donor to recipient bacterium by
- (1) Conjugation tube
 - (2) Bacteriophage
 - (3) Fungi
 - (4) Prions
103. During conjugation between F^+ and F^- bacterial cells the F^- cell is converted to
- (1) F^+
 - (2) Hfr
 - (3) F^-
 - (4) Meta male

- 104.** Select the incorrect option w.r.t. binary fission method in bacteria shown below in the diagram.



- (1) Occurs under favourable condition
 - (2) Involves DNA replication
 - (3) Sort of sexual reproduction
 - (4) The daughter cells formed are genetically identical
- 105.** Characteristic photosynthetic pigments present in BGA are
- (1) Chlorophyll a, Chlorophyll b
 - (2) Chlorophyll a, phycocyanin and phycoerythrin
 - (3) Chlorophyll c, Phycobilins
 - (4) Chlorophyll c, phycocyanin and phycoerythrin
- 106.** Which of the following is not related to BGA?
- (1) Nitrogen Fixation
 - (2) Cyanobacteria
 - (3) Oxygenic photosynthesis
 - (4) Nonoxygenic photosynthesis
- 107.** Which of the following played a significant role in the evolution of O_2 on earth?
- (1) Archaeobacteria
 - (2) BGA
 - (3) Green Algae
 - (4) Protistan algae
- 108.** Edible fast growing cyanobacteria used as source of single cell protein (SCP) is
- (1) *Spirulina*
 - (2) *Nostoc*
 - (3) *Anabaena*
 - (4) *Microcystis*
- 109.** Algal bloom in the polluted water bodies are often formed by
- (1) Blue green algae
 - (2) Sulphur bacteria
 - (3) Nitrifying bacteria
 - (4) Archaeobacteria

- 110.** Which of the following organisms are responsible for the production of biogas from the dung of cows and buffaloes?
- (1) Methanogens
 - (2) Halophiles
 - (3) Thermoacidophiles
 - (4) Actinomycetes
- 111.** Organisms with soap box like structure are
- (1) Diatoms
 - (2) Dinoflagellates
 - (3) Consumer decomposer protist
 - (4) Euglenoids
- 112.** A longitudinal and a transverse flagellum are feature of organisms
- (1) Diatoms
 - (2) Sporozoans
 - (3) Slime moulds
 - (4) Dinoflagellates
- 113.** Which organisms undergo such a rapid multiplication that they make the sea appear red (called red tides)?
- (1) *Navicula*
 - (2) *Trichodesmium*
 - (3) *Gonyaulax*
 - (4) *Chlamydomonas*
- 114.** Cell wall is indestructible in diatoms due to presence of
- | | |
|------------|---------------|
| (1) Pectin | (2) Cellulose |
| (3) Silica | (4) Chitin |
- 115.** Diatomaceous earth is used for all except
- (1) Polishing
 - (2) Filtration of oil and syrup
 - (3) Conductor in refrigerator
 - (4) Soundproofing
- 116.** are the chief producers in the oceans.
- (1) Diatoms
 - (2) Dinoflagellates
 - (3) BGA
 - (4) Euglenoids

- 117.** Viruses were called as *contagium vivum fluidum*
(1) Stanley (2) Iwanowsky
(3) Beijerinck (4) Pasteur
- 118.** Viroids are small infectious particles which are lack of
A. Capsid B. Enzyme
C. RNA D. DNA
(1) A, B and D (2) B and D
(3) A, B and C (4) A and D
- 119.** The protein coat of virus is made of subunits called
(1) Peplomere (2) Capsid
(3) Capsomere (4) Shell
- 120.** Bacteria reproduce by binary fission during A and endospore formation during B condition

A	B
(1) Favourable	Favourable
(2) Unfavourable	Favourable
(3) Favourable	Unfavourable
(4) Unfavourable	Unfavourable
- 121.** Endospores are
(1) Thin walled and Metabolically inactive
(2) Thick walled and stress intolerant
(3) Thick walled and resistant
(4) Metabolically active reproductive structure of bacteria
- 122.** Select the incorrect match w.r.t. reproduction in bacteria.
(1) Transformation - Griffith
(2) Conjugation - Done without F-Plasmid
(3) Endospore formation - Reproduction without multiplication
(4) Binary fission - Most common method of multiplication
- 123.** Cyanobacteria do not possess/ show
(1) Flagella
(2) Gene recombination
(3) 70S Ribosome
(4) Lamellosome
- 124.** Heterocyst present in cyanobacteria is specialised for N_2 fixation and it possess
(1) Thick walled (2) Oxygen
(3) Nitrogenase (4) Both (1) and (3)
- 125.** Which is true for archaeobacteria?
(1) All are Methanogens
(2) All are chemoautotroph
(3) Oldest living fossils
(4) All are fossils
- 126.** In five kingdom classification, multicellular Eukaryotic organism with autotrophic mode of nutrition are included in
(A) Animalia (B) Fungi
(C) Plantae (D) Protista
(1) Only (C) is correct
(2) Both (A) and (B) are correct
(3) Only (B) is correct
(4) All are correct
- 127.** Free living nitrogen fixing bacteria are found in
(1) Air (2) Soil
(3) Root nodules (4) All of these
- 128.** The function of mesosome in prokaryotes
(1) Respiration (2) DNA replication
(3) Both (1) and (2) (4) Nitrogen fixation
- 129.** Which of the following can utilize light energy for the synthesis of ATP
(1) Slime mould
(2) Cephaleuros
(3) Halophiles
(4) Thermoacidophiles
- 130.** Harmful activity of BGA is :-
(1) N_2 -Fixation
(2) Increase fertility of soil
(3) Water bloom
(4) Increase alkalinity of soil
- 131.** In bacteria respiratory enzymes are present in
(1) Cell wall (2) Mitochondria
(3) Plasma membrane (4) Nucleoid
- 132.** On the basis of their shape bacteria are grouped under _____ categories.
(1) Two (2) Three (3) Six (4) Four
- 133.** Example of methanogen archaeobacteria
(1) Methanococcus (2) BGA
(3) Slime mould (4) Mycoplasma
- 134.** Archaeobacteria found in hot springs are known as
(1) Methanogens
(2) Thermoacidophiles
(3) Cryophiles
(4) Halophiles
- 135.** Mycoplasma is not affected by penicillin due to
(1) Pleomorphic nature
(2) Presence of linear DNA
(3) Lack of cell wall
(4) Presence of peptidoglycan cell wall

- | | |
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| <p>136. <i>Pleurobrachia</i> is called</p> <ol style="list-style-type: none"> (1) Sea walnut (2) Sea gooseberry (3) Jellyfish (4) Sea anemone <p>137. Tentacles are</p> <ol style="list-style-type: none"> (1) Present in <i>Beroe</i> but absent in <i>Ctenoplana</i> (2) Present in <i>Ctenoplana</i> but absent in <i>Beroe</i> (3) Absent in both <i>Beroe</i> and <i>Ctenoplana</i> (4) Present in both <i>Beroe</i> and <i>Ctenoplana</i> <p>138. In Ctenophora, cilia are present to form</p> <ol style="list-style-type: none"> (1) Four median comb plates (2) Eight median comb plates (3) Ten median comb plates (4) Six median comb plates <p>139. Which of the following is not a common character for both Coelenterata and Ctenophora?</p> <ol style="list-style-type: none"> (1) Cnidoblast (2) Diploblastic (3) Tissue grade organization (4) Nerve cell without controlling centre <p>140. Life cycle of <i>Pleurobranchia</i> includes which of the following larva?</p> <ol style="list-style-type: none"> (1) Scyphistoma (2) Miracidium (3) Dipleura (4) Cydippid <p>141. Which of the following is incorrect w.r.t. Ctenophora?</p> <ol style="list-style-type: none"> (1) Exclusively marine (2) Bioluminescent (3) Dioecious (4) Digestion both extracellular and intracellular <p>142. Ctenophores are also called</p> <ol style="list-style-type: none"> (1) Cnidarians (2) Sponges (3) Comb jellies (4) Flatworms | <p>143. Coelom in flat worms is</p> <ol style="list-style-type: none"> (1) Poorly developed (2) Absent (3) Schizocoel (4) Haemocoelom <p>144. Dog tapeworm is</p> <ol style="list-style-type: none"> (1) <i>Dracunculus</i> (2) <i>Taenia</i> (3) <i>Schistosoma</i> (4) <i>Echinococcus</i> <p>145. Beef tapeworm is</p> <ol style="list-style-type: none"> (1) <i>Taenia saginata</i> (2) <i>Taenia solium</i> (3) <i>Schistosoma</i> (4) <i>Dracunculus medinensis</i> <p>146. Flatworm have</p> <ol style="list-style-type: none"> (1) Organ level of organization (2) Ladder like nervous system (3) Triploblastic body (4) All of these <p>147. Which of the following phylum includes bilaterally symmetrical, triploblastic and acoelomate animals?</p> <ol style="list-style-type: none"> (1) Ctenophora (2) Platyhelminthes (3) Aschelminthes (4) Cnidaria <p>148. Mark the odd one w.r.t. larval stages of tapeworm.</p> <ol style="list-style-type: none"> (1) Cercaria (2) Onchosphere (3) Cysticercus (4) Bladderworm <p>149. Which of the larval stages are absent in blood fluke?</p> <ol style="list-style-type: none"> (1) Miracidium and sporocyst (2) Redia and metacercaria (3) Sporocyst and redia (4) Cercaria and metacercaria |
|---|---|

150. Which of the following is the **correct** sequence of larval stages in liver fluke?
- (1) Miracidium → Redia → Sporocyst → Cercaria
(2) Miracidium → Sporocyst → Redia → Cercaria
(3) Sporocyst → Miracidium → Redia → Cercaria
(4) Redia → Sporocyst → Miracidium → Cercaria
151. *Wuchereria bancrofti* is transmitted by :-
- (1) Anopheles mosquito
(2) Culex mosquito
(3) Tse-tse fly
(4) None of these
152. Body cavity of an adult *Ascaris* is called :-
- (1) Coelom
(2) Enterocoel
(3) Haemocoel
(4) Pseudocoel
153. Female *Ascaris* differs from the male in having
- (1) Curved tail (2) Pineal setae
(3) Longer body (4) None of these
154. Which of the following is called eye worm?
- (1) *Loa loa* (2) *Dracunculus*
(3) *Ancylostoma* (4) *Enterobius*
155. Endoparasite of small intestine of humans is
- (1) *Wuchereria* (2) *Ancylostoma*
(3) *Ascaris* (4) Both (2) and (3)
156. Amphids are chemoreceptor found in
- (1) *Taenia solium* (2) *Ascaris*
(3) *Euplectella* (4) *Aurelia*
157. Mark the correct statement regarding the organism in figure.



- (1) Male are longer than female
(2) Male possess a curved tail

- (3) Female are longer than male
(4) Both (2) and (3)
158. Which of the following is not an intestinal parasite?
- (1) *Taenia* (2) *Trichuris*
(3) *Ancylostoma* (4) *Wuchereria*
159. Setae are present in all, except
- (1) *Pheretima* (2) *Nereis*
(3) *Lumbricus* (4) *Hirudinaria*
160. In annelids, respiration takes place through
- (1) Trachea (2) Book-lungs
(3) Moist skin (4) Lungs
161. The excretory structures in annelids are
- (1) Malpighian tubules
(2) Renette cells
(3) Nephridia
(4) Solenocytes
162. The saliva of _____ contains an anticoagulant hirudin
- (1) Leech (2) Earthworm
(3) Sea mouse (4) Sandworm
163. All of the following are characters of annelids, except
- (1) Metameric segmentation
(2) Cutaneous respiration
(3) Complete digestive tract
(4) Pseudocoelom
164. In earthworm, the first body segment is called
- (1) Peristomium (2) Prostomium
(3) Pygidium (4) None of these
165. In earthworm:
- (1) Development is direct
(2) Setae are present
(3) Clitellum is permanent
(4) All are correct
166. Which respires by book lungs?
- (1) Insect (2) Scorpion
(3) King crab (4) Prawn
167. In arthropods the body cavity is called
- (1) Haemocoel (2) Heterocoel
(3) Enteron (4) None of these
168. Mouth parts of housefly are of

- | | |
|--|---|
| <p>(1) Siphoning
(2) Sponging
(3) Piercing and sucking
(4) Chewing and lapping</p> <p>169. In honey bee, drones are
(1) Haploid females (2) Worker bees
(3) Haploid males (4) Diploid male</p> <p>170. The connecting link between annelids and arthropods is
(1) <i>Limulus</i> (2) Cockroach
(3) <i>Peripatus</i> (4) <i>Neopilina</i></p> <p>171. Wiggler is the larva of
(1) Mosquito (2) Butterfly
(3) Housefly (4) Sandfly</p> <p>172. Phylum Arthropoda is characterized by
(1) Unjointed appendages
(2) Absence of appendages
(3) Jointed appendages
(4) None of these</p> <p>173. The vector of <i>Wuchereria</i> is
(1) <i>Culex</i> (2) <i>Anopheles</i>
(3) <i>Aedes</i> (4) <i>Triatoma</i></p> <p>174. <i>Neopilina</i> is a connecting link between
(1) Platyhelminthes and aschelminthes
(2) Mollusca and helminthes
(3) Arthropoda and Mollusca
(4) Annelida and Mollusca</p> | <p>175. Trochophore larva occurs in
(1) Aschelminthes and annelids
(2) Coelenterata and annelida
(3) Annelida and porifera
(4) Annelida and Mollusca</p> <p>176. Name the branch of science dealing with Mollusca
(1) Entomology (2) Parasitology
(3) Malacology (4) Saurology</p> <p>177. Study of molluscan shells is
(1) Malacology (2) Entomology
(3) Conchology (4) Acarology</p> <p>178. Organs of Bojanus are
(1) Excretory organs of <i>unio</i>
(2) Endocrine organs of insects
(3) Respiratory organs of star fish
(4) Chemoreceptor of <i>peripatus</i></p> <p>179. In which members of phylum, the body is organized into head, foot and visceral hump?
(1) Arthropoda
(2) Echinodermata
(3) Mollusca
(4) Annelida</p> <p>180. The blood of molluscs contains amoebocytes and a copper containing blue respiratory pigment called
(1) Pinnaglobin
(2) Haemocyanin
(3) Haemoglobin
(4) Chlorocruonin</p> |
|--|---|



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AIR - 35  AVNIKA JAIN Distance	AIR - 37  LOKESH MANDOLECHA Distance	AIR - 38  VIRENDRA CHOUDHARY Distance	AIR - 41  ARJUN SARASWAT Distance	AIR - 42  NISHTHA AGRAWAL Classroom	AIR - 43  SANKALPA KESHARI Distance	AIR - 47  PARUL BERRY Classroom	AIR - 48  SAMAGRA SINGH Classroom	AIR - 49  SAGAR MISHRA Classroom	AIR - 50  AARUSHI KOLLANA VISHWESH M. BHARADIYA Distance	AIR - 51  PRATHAM SINGLA Distance	AIR - 52  RAGHAVEN KHANDAL Distance	AIR - 55  AYUSHI MITTAL Classroom	AIR - 58  AYUSHI BANSAL Classroom	AIR - 60  AYUSHI BANSAL Classroom
AIR - 62  RIYA SINGH Classroom	AIR - 65  PRATEEK K. MISHRA Classroom	AIR - 66  VAIBHAV MITTAL Classroom	AIR - 69  VAIBHAV GARG Distance	AIR - 70  AYAN AGARWAL Distance	AIR - 73  SANJEEV REDDY Distance	AIR - 74  SAVI GADEONE Distance	AIR - 75  PRANJAL PRASAD Distance	AIR - 77  PRAGATI AHUJA Distance	AIR - 78  PARVA N. LADANI Distance	AIR - 79  PULKIT GUPTA Classroom	AIR - 81  V.N.T.V. BELLAMKONDA Distance	AIR - 83  PRABHU SINGH Classroom	AIR - 84  RUTUDHRAJ SAVALIYA Classroom	AIR - 90  ESHVARK AGGARWAL Distance
AIR - 91  SHUBHAM MARODIA Distance	AIR - 95  KSHITIJ AGARWAL Classroom	81230* Classroom 50267 Distance 30963										AIR - 96  JITENDRA YADAV Classroom	AIR - 100  VIDUSHI VARSINEY Classroom	

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